AI Resume Analyzer & Job Matcher

GitHub README Documentation

About the Project

An intelligent resume analysis web application built using **Python and Streamlit** that helps users analyze their resumes, compare skills with job descriptions, identify missing skills, and discover suitable job roles.

The application is designed to simplify the job application process. Users can upload a resume in PDF format, enter a job description, and receive insights into resume-job compatibility and potential skill gaps.

Features

- **Resume Upload:** Upload resumes in PDF format.
- **Job Description Analysis:** Enter a job description for comparison.
- **Resume Match Score:** Calculate a resume-job matching percentage.
- **Missing Skills Detection:** Identify skills that may be missing from the resume.
- **Job Role Recommendations:** Suggest suitable roles based on extracted skills.
- **Interactive Interface:** Use an easy-to-navigate Streamlit web interface.

Technologies Used

- **Python** — core programming language.
- **Streamlit** — web application framework.
- **Pandas** — data manipulation and analysis, if used in the implementation.
- **PDF processing library** — extracts text from uploaded PDF resumes.

Project Structure

```
AI_Resume_Analyzer/
■■■ app.py                # Main Streamlit application
■■■ resume_parser.py      # Resume text extraction
■■■ matcher.py            # Resume-job matching logic
■■■ skills.py             # Skill extraction and analysis
■■■ requirements.txt      # Python dependencies
■■■ README.md             # Project documentation
```

Installation and Setup

1. Clone the repository

```
git clone YOUR_GITHUB_REPOSITORY_URL
```

Replace `YOUR_GITHUB_REPOSITORY_URL` with your actual GitHub repository URL.

2. Navigate to the project directory

```
cd AI_Resume_Analyzer
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Run the application

```
streamlit run app.py
```

5. Open the application

Streamlit will show a local URL, usually `http://localhost:8501`. Open it in your browser.

Deployment

The application can be deployed using Streamlit Community Cloud:

- Push the project files to a GitHub repository.
- Open Streamlit Community Cloud.
- Connect your GitHub repository.
- Select `app.py` as the main application file.
- Click **Deploy** and wait for the application to build.

After deployment, Streamlit provides a URL for accessing the application online.

Project Objectives

- Simplify resume analysis for job seekers.
- Compare candidate skills with job requirements.
- Identify potential skill gaps.
- Help candidates explore suitable job opportunities.
- Provide a simple interface for resume evaluation.

Future Enhancements

- Integrate NLP techniques for more advanced resume analysis.
- Provide detailed resume analysis reports.
- Recommend courses to improve missing skills.
- Support additional resume formats.
- Improve job recommendations using semantic matching.

Developed By

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This project is developed for educational and portfolio purposes.

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